



Give Bit the Steps

Quick facilitation notes & answer key

Goal

Children compose a short arrow program that moves Bit to the star — choosing the direction and the number of steps. Writing the steps that solve a problem is composing a simple algorithm (the create level of sequencing).

Kindergarten frame (Ontario)

Problem Solving and Innovating — children create a sequence of directions to reach a goal.

No reading required of the child: read each prompt aloud; the child draws ➡/➡ arrows in the boxes.

Materials

The printed worksheet + a pencil. Optional: floor boxes so children walk and then record the steps they took.

Run it (about 15 min)

1. MODEL: box 1 to the star on box 3 — 'two boxes right, so ➡ ➡.' Count the gap, then write that many arrows.
2. GUIDED: Exercise 1 — box 1 to box 4 is three boxes right ➡ ➡ ➡.
3. YOU TRY: one step (Ex 2 ➡ ➡), then a left move (Ex 3 ➡ ➡).
4. Connect: 'Writing the steps that solve a problem is making an algorithm — that is what coders do.'

Answer key (modelled by the run-gate)

Model — box 1 ➡ star box 3: ➡ ➡.

Ex 1 — box 1 ➡ box 4: ➡ ➡ ➡ (three boxes right).

Ex 2 — box 2 ➡ box 3: ➡ (one box right).

Ex 3 — box 4 ➡ box 2: ➡ ➡ (two boxes left).

Watch for & differentiate

Common slip: too many or too few arrows — count the boxes between Bit and the star first.

Simplify: do only right-moving goals before introducing the left move.

Extend: place the star farther away (box 5) and write the longer program.

Movement option: a child walks the route and the class records one arrow per step.

Success looks like

The child draws the correct number of arrows in the correct direction so the program reaches the star.